

Fractions — Answer Key

#	Item ID	Correct	Target node(s)	q_vector
1	MULT_001	C (42)	Multiplication	1000000000
2	MULT_002	C (450)	Multiplication	1000000000
3	DIV_001	C (4 each, 1 left over)	Division	0100000000
4	DIV_002	B (0.5)	Division	0100000000
5	CONC_001	B (2/5)	Concept_of_fraction	0010000000
6	CONC_002	B (Kofi only)	Concept_of_fraction	0010000000
7	EQV_001	C (4/6)	Equivalent_fractions	1011000000
8	EQV_002	B (9)	Equivalent_fractions	1011000000
9	SIMP_001	B (2/3)	Simplifying_fractions	0111100000
10	SIMP_002	B (2/3)	Simplifying_fractions	0111100000
11	MTI_001	D (7/3)	Mixed_to_improper	1010010000
12	MTI_002	D (15/4)	Mixed_to_improper	1010010000
13	ITM_001	B (2 3/4)	Improper_to_mixed	0110001000
14	ITM_002	B (3 2/5)	Improper_to_mixed	0110001000
15	MUF_001	C (8/15)	Multiplying_fractions	1010000100
16	MUF_002	C (6/20)	Multiplying_fractions	1010000100
17	DIF_001	C (6/4)	Dividing_fractions	1010000110
18	DIF_002	B (3)	Dividing_fractions	1010000110
19	PED_001	D (4/3)	PEDMAS	1110000111
20	PED_002	A (1)	PEDMAS	1110000111
21	MULTI_001	B (7/6)	Mixed_to_improper, Multiplying_fractions	1010010100
22	MULTI_002	B (2/3)	Equivalent_fractions, Simplifying_fractions	1111100000
23	MULTI_003	A (1/4)	Multiplying_fractions, Simplifying_fractions	1111100100
24	MULTI_004	C (4)	Mixed_to_improper, Multiplying_fractions, Dividing_fractions, PEDMAS	1110010110
25	MULTI_005	B (3 3/4)	Mixed_to_improper, Improper_to_mixed, Multiplying_fractions	1110011100
26	MULTI_006	C (4/3)	Mixed_to_improper, Multiplying_fractions, Dividing_fractions, PEDMAS	1110010110

Node order in q_vector: Multiplication, Division, Concept_of_fraction, Equivalent_fractions, Simplifying_fractions, Mixed_to_improper, Improper_to_mixed, Multiplying_fractions, Dividing_fractions, PEDMAS

How to score: mark each item right or wrong, build a response vector across all items in order, then run NPC or DINA-MLE classification against the feasible knowledge states derived from the DAG.